CONTACT PROCESS ON SCALE-FREE PERCOLATION IN CONTINUUM SPACE

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Introduction

Motivation:

Commercial movements of cattle form a large and complex network, serving as a significant vector for the spread of infectious diseases.

The scale-free percolation model (SFP), characterized by its **scale-free property**, **positive clustering coefficient**, and **spatial component**, is particularly suitable for modeling this network.

Objective:

1. Study the behavior of a toy epidemic model on SFP random graph in continuum space.
2. Parameter inference on SFP to fit real dataset.
3. Develop a generative model to create a more accurate and dynamic tool for predicting epidemic spread on this network.

1. Scale-free percolation model (SFP)

- **Vertices** are given by a Poisson point process $\mathcal{X}$ on $\mathbb{R}^d$ with intensity 1.
- **Random weights**, W_x are assigned to each vertex $x \in \mathcal{X}$, with power law distribution of exponent $\tau > 1$, i.e.

$$\mathbb{P}(W_x \geq t) = \ell(t)\, t^{-(\tau-1)}, \quad \text{for } t > 1,$$

for ℓ a slowly-varying function at infinity.

- **Edges** between pairs of vertices $x, y \in \mathcal{X}$ are drawn, conditionally on their weights W_x and W_y, with probability given by

$$p_{x,y} = 1 - \exp\left(-\rho \frac{W_x W_y}{||x-y||^{\alpha}}\right),$$

where $\rho > 0$, $\alpha > 0$ and $||\cdot||$ is the Euclidean norm.

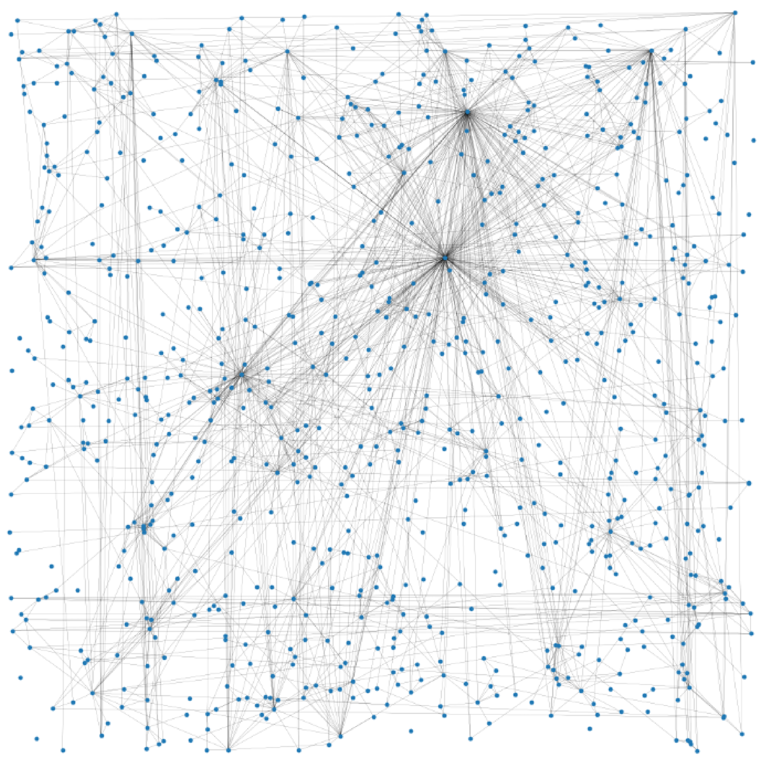

Figure 1. SFP simulation for $\tau = 2.5$ and $\alpha = 1.8$.

Parameter set:

- $d > 1$: dimension of space.
- $\tau > 1$: parameter representing the graph's inhomogeneity.
- $\alpha > 0$: long-range parameter.
- $\rho > 0$: percolation parameter.

2. Features of SFP

Scale-free property: (Deprez and Wüthrich 2019)

If D_x is the degree of a vertex x chosen uniformly at random, then D_x follows a power law distribution of parameter $\gamma = \frac{\alpha}{d}(\tau - 1)$, i.e.

$$\mathbb{P}(D_x > k) = \ell(k) k^{-(\gamma-1)},$$

for ℓ a slowly-varying function at infinity.

Percolation and graph distance: (Deprez and Wüthrich 2019)

Let ρ_c be the critical percolation parameter defined as

$$\rho_c = \inf\{\rho,\ \mathbb{P}_0\left[|\mathcal{C}(0)| = \infty\right] > 0\},$$

where the probability $\mathbb{P}_0$ is the Palm version of $\mathbb{P}$ conditioned on having a vertex in the origin, and $\mathcal{C}(0)$ is the connected component of the origin.

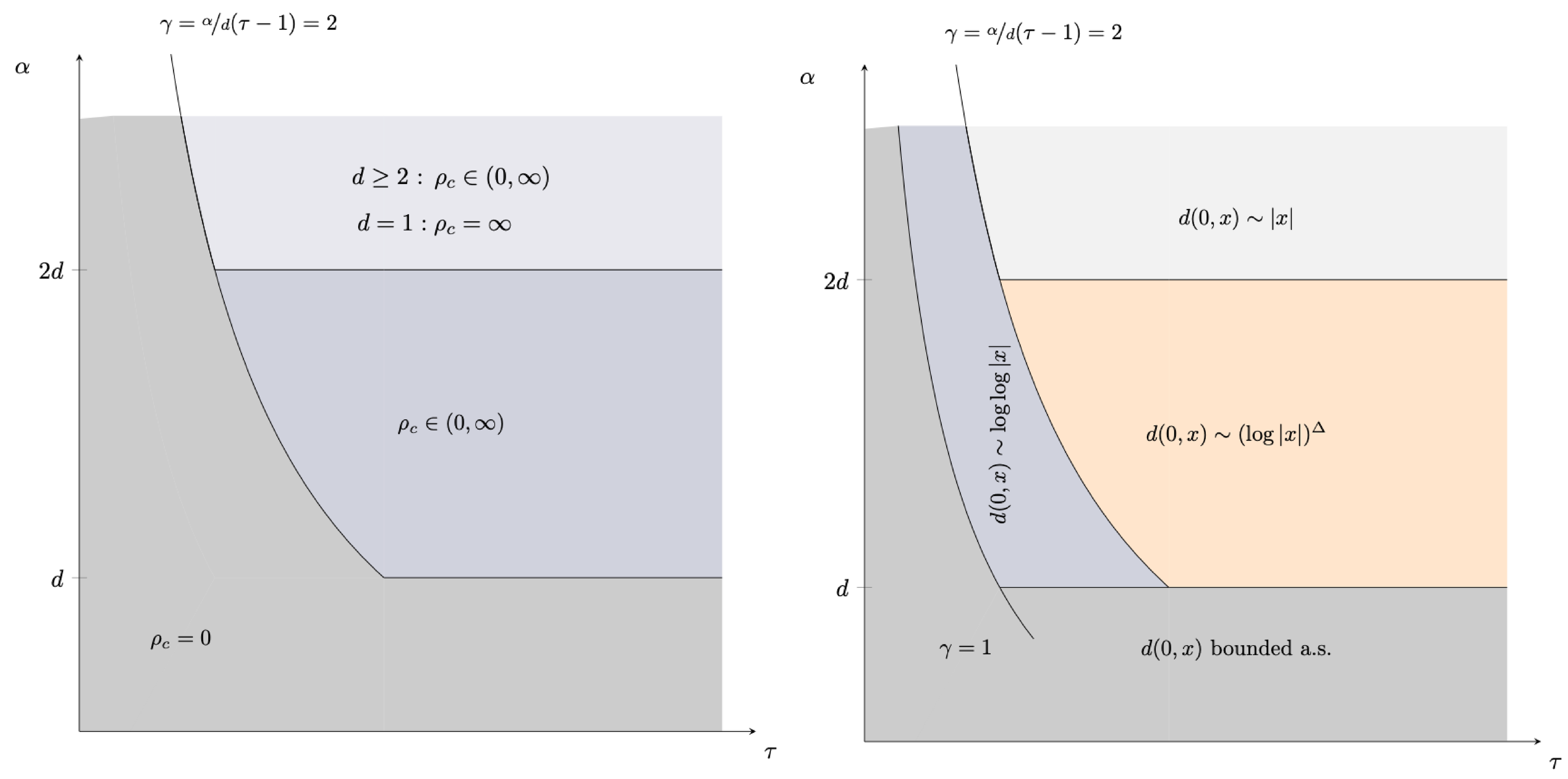

Figure 2. Phase transition (lhs) and graph distances (rhs) in SFP in $\mathbb{R}^d$, $d > 1$, for different values of α and τ.

References

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3. Ultra small-world regime ($\gamma \in (1,2)$)

- Let $(\xi_t)_t$ be the contact process, an SIS model with infection rate λ and recovery 1.
The *non-extinction probability* of the contact process starting at the origin is defined as follows

$$\Gamma(\lambda) = \mathbb{P}\left(\xi_t^0 \neq \emptyset, \forall t \geq 0\right),$$

where (ξ_t^0) is the contact process starting from only the origin infected.

Theorem 1: Non-extinction probability of the CP

As $\lambda \to 0$,

$$\Gamma(\lambda) \asymp \begin{cases} \lambda^{\frac{1}{2-\gamma}}, & \text{if } \gamma \in (1, \frac{3}{2}] \\ \frac{\lambda^{2\gamma-1}}{\log(1/\lambda)^{\gamma-1}} & \text{if } \gamma \in (\frac{3}{2}, 2] \end{cases}$$

- Let $\mathcal{G}_n$ be the restriction of SFP to vertices in $\left[0, n^{1/d}\right)^d$.
The *extinction time* of the contact process starting from full occupancy is defined as follows

$$\tau_{\mathcal{G}_n} = \inf\{t : \xi_t^{\mathcal{G}_n} = \emptyset\},$$

where $(\xi_t^{\mathcal{G}_n})_t$ is the contact process starting from the graph fully infected.

Theorem 2: Metastability of the CP on SFP

For every $\lambda > 0$, *there exists a* $c > 0$ *such that*

$$\mathbb{P}\left(\tau_{\mathcal{G}_n} \geq e^{cn}\right) \to 1, \quad \text{as } n \longrightarrow \infty.$$

4. Small-world regime ($\gamma > 2$, $\rho > \rho_c$, $\alpha \in (d, 2d)$)

Theorem 3: Extinction time of the CP on SPF (Work in progress)

For every $\lambda > 0$, *there is a constant* $c > 0$ *such that for every constant* A *such that* $A > \gamma + 1$,

$$\mathbb{P}\left(\tau_{G_n} \geq \exp\left(c \frac{n}{(\log n)^A}\right)\right) \longrightarrow 1, \quad \text{as } n \longrightarrow \infty.$$

5.Proofs

Main tool: Scale-free property and controlled graph distance allow the existence of a chain of hubs where the epidemic persists.

Theorem 1: (Inspired by Linker et al. 2021)

The main idea of the proof consists on analyzing the probability of the epidemic reaching a chain of powerful vertices, allowing the conservation of the epidemic.

Extinction time:

Mountford et al. 2016 state that if there is a sub-graph containing N stars with at least S neighbors and distanced by maximum D vertices such that

$$S \geq \lambda^{-2}, \quad S > \lambda^{-2} \log\left(\frac{1}{\lambda}\right) D,$$

then

$$\mathbb{P}(\tau_{\mathcal{G}_n} \geq e^{c(\lambda^2 S + N)}) \to 1.$$

Theorem 2: (Inspired by Linker et al. 2021)

We show the existence of such a sub-graph where S and D are independent of n and $N = O(n)$.

Theorem 3:

In this proof, S and D are dependent of n.
With $S = \log n$, $D = \log\log n$ and $N = O\left(\frac{n}{(\log n)^A}\right)$, we conclude.

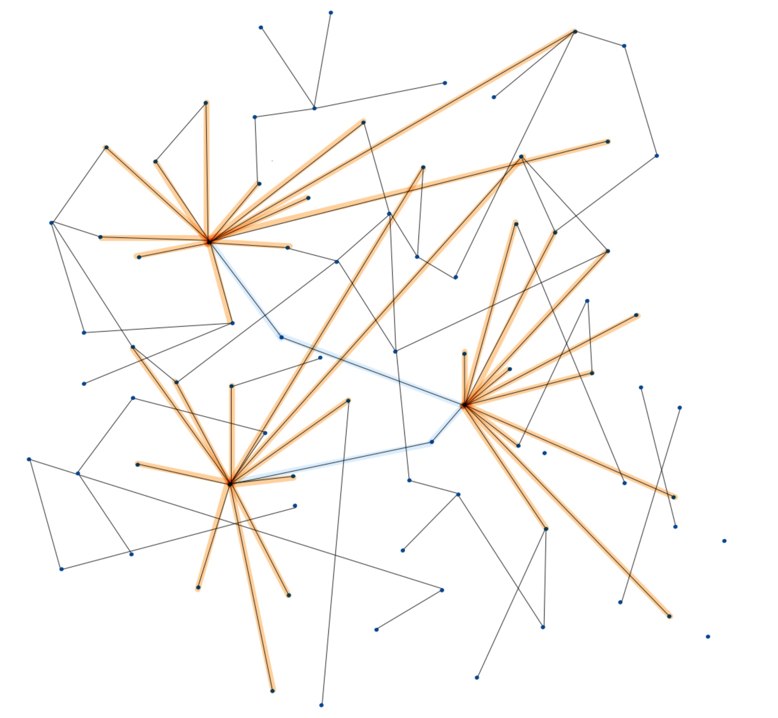

Figure 3. Toy example of sub-graph where $N = 3$, $S = 11$ and $D = 2$.

6. Perspectives

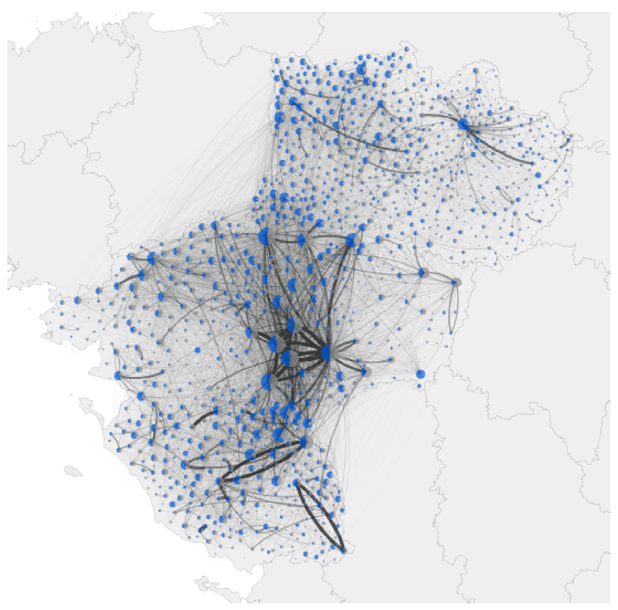

Figure 4. Representation of the BDNI (Base de Données Nationale d'Interactions) of the Pays de la Loire region between 2010 and 2014. (Graphic by Gaël Beaunée).

- Parameter inference on τ and α to fit French cattle identification database (BDNI) using *Approximate Bayesian Computation (ABC)*.

Advantages of ABC methods:
  - Independent of likelihood function
  - Simulation based method

- Develop a generative model representing the French cattle movement network.
- Study the spread of complex epidemic models on this network.